Dear **Data Analytics & Visualization** group,

I am a **Computational Physicist** (Double Degree in Physics & Mathematics, MSc in Plasma Physics) and **HPC Scientific Software Engineer** in BSC's Performance Team. Daily, I deploy, couple, fix and optimize simulation workloads across different clusters. But I am more than what those roles can tell about me, I want to create, to be able to take the raw material and make it malleable to the public, to tell stories.

When creativity kicks in, I use my free time to do some independent creative computing. The idea is turning technical ideas into tangible realities:

- **Music engine for MN5** (MNusic) — A real-time music engine transforming live cluster telemetry into generative harmonic soundscapes and reactive particle fields. Much inspired by Brian Eno's works, I wondered how visitors could hear the pulse of computational science.
- **Cat orchestra** (Pipa-Music-Ball) — While away from home, I always wondered what my cat Pipa was doing. I had this thought of using a microcontroller to capture movement data of a toy she liked, then mapping feline motion vectors to microtonal audio scales, proving that sensor physicalization can be tactile and playful.
- **Preparing an exhibition** (Expo-Curator) — I was contacted to do a photo exhibition for the first time and this drove me crazy. Creating beautiful compositions out of thousands of photos becomes a difficult art. I created this open-source gallery staging platform automating artwork classification, catalog layout, and exhibition wall geometry for scientific dissemination spaces that helps you focus on what matters, the photos!

Treating **LLMs and agentic pair-programming as fluid thinking partners** allows me to transition without friction between C++/Fortran HPC kernels, microcontroller firmware, browser Web Audio synthesis, and Adobe After Effects procedural visual essays.

Being grateful to have followed some of your workshops and tracked your work, whether it is **scientific storytelling, interactive visualization or creative intelligence**, I believe my background and skills put me in sweet intersection where my computational physics grounding, digital twin workflows, and sensory media experiments belong.

It would be lovely to, at some point, attend your workshops not from the audience but presenting a new idea from the Creative Intelligence Lab or to co-engineer data installations for the public to enjoy.

Sincerely and with great enthusiasm,
Carlos Romero Madrid

✦ LIVE INTERACTIVE WEB DOSSIER & DIGITAL EXPERIENCES
If you are now interested in my CV, I created this online version for the DAV group, hope you like it! → https://carrommad.github.io/cool-cv